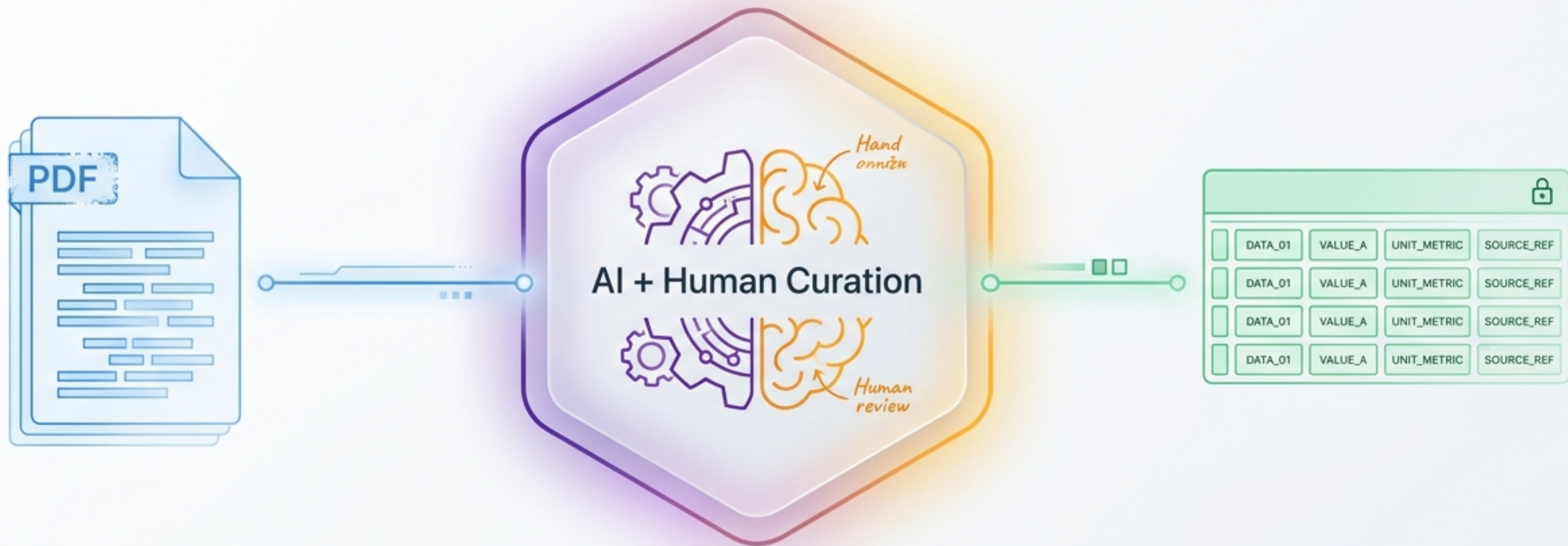




Papers-to-Table: Auditable Scientific Data Extraction

A local-first system for transforming scientific PDFs into structured tables via AI-assisted human curation.

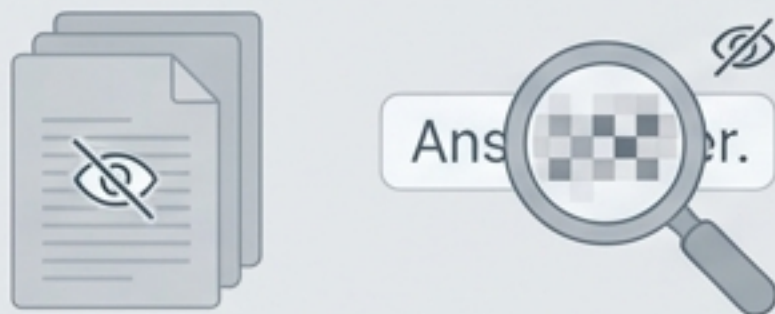
Chat Is Not Curation

Generic PDF Chat

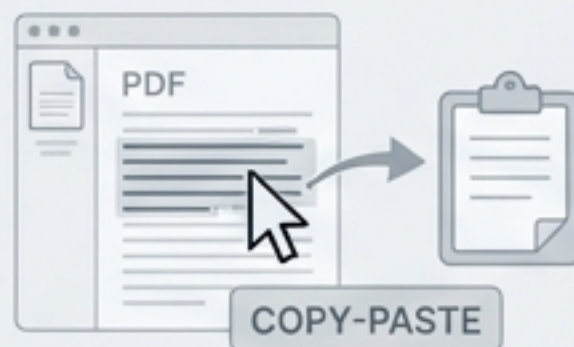
Workflow:
Open-ended,
conversational
exploration.



Evidence:
Hidden
provenance;
opaque AI guesses.



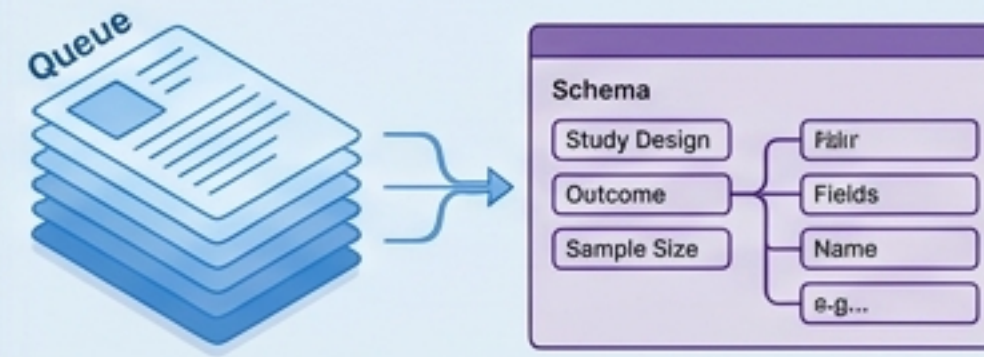
Execution:
Manual
copy-pasting
to external tables.



The Golden Rule:
Human review is
structurally required.
AI proposes, humans decide.

Papers-to-Table

Workflow:
Queue-first,
schema-driven
alignment.



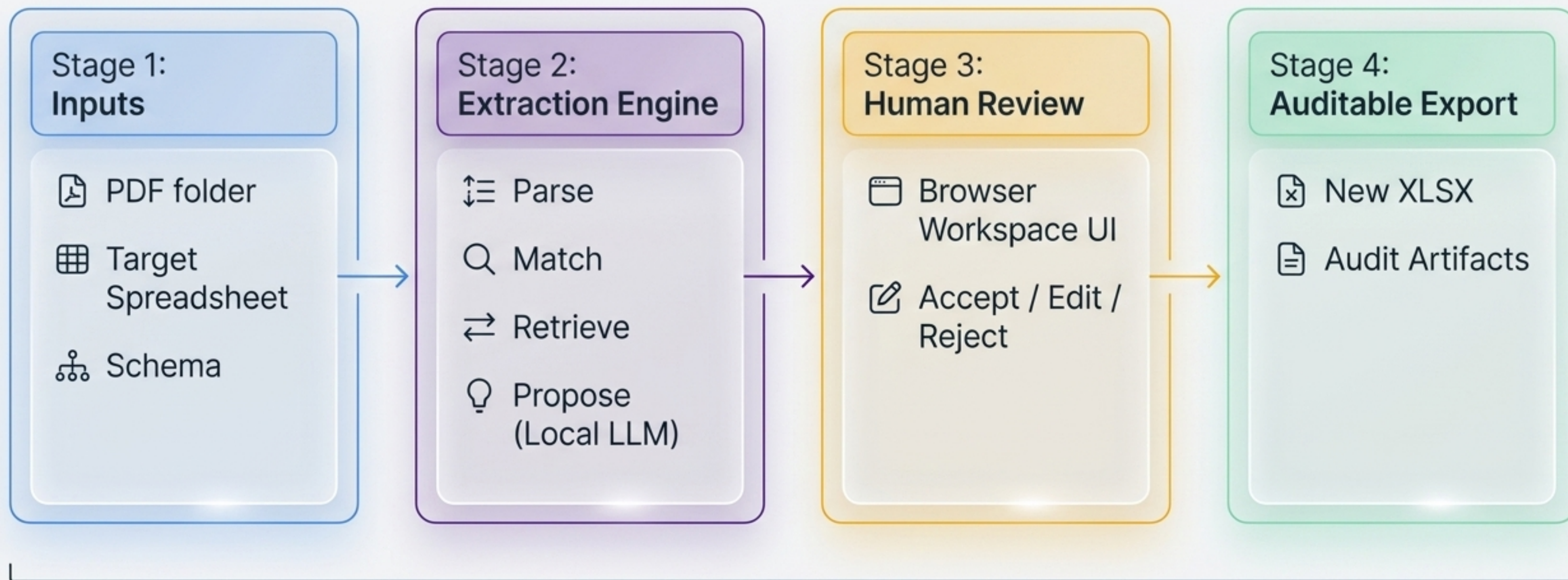
Evidence:
Auditable
provenance; exact
source quotes.



Execution:
Strict export
of accepted
changes only.

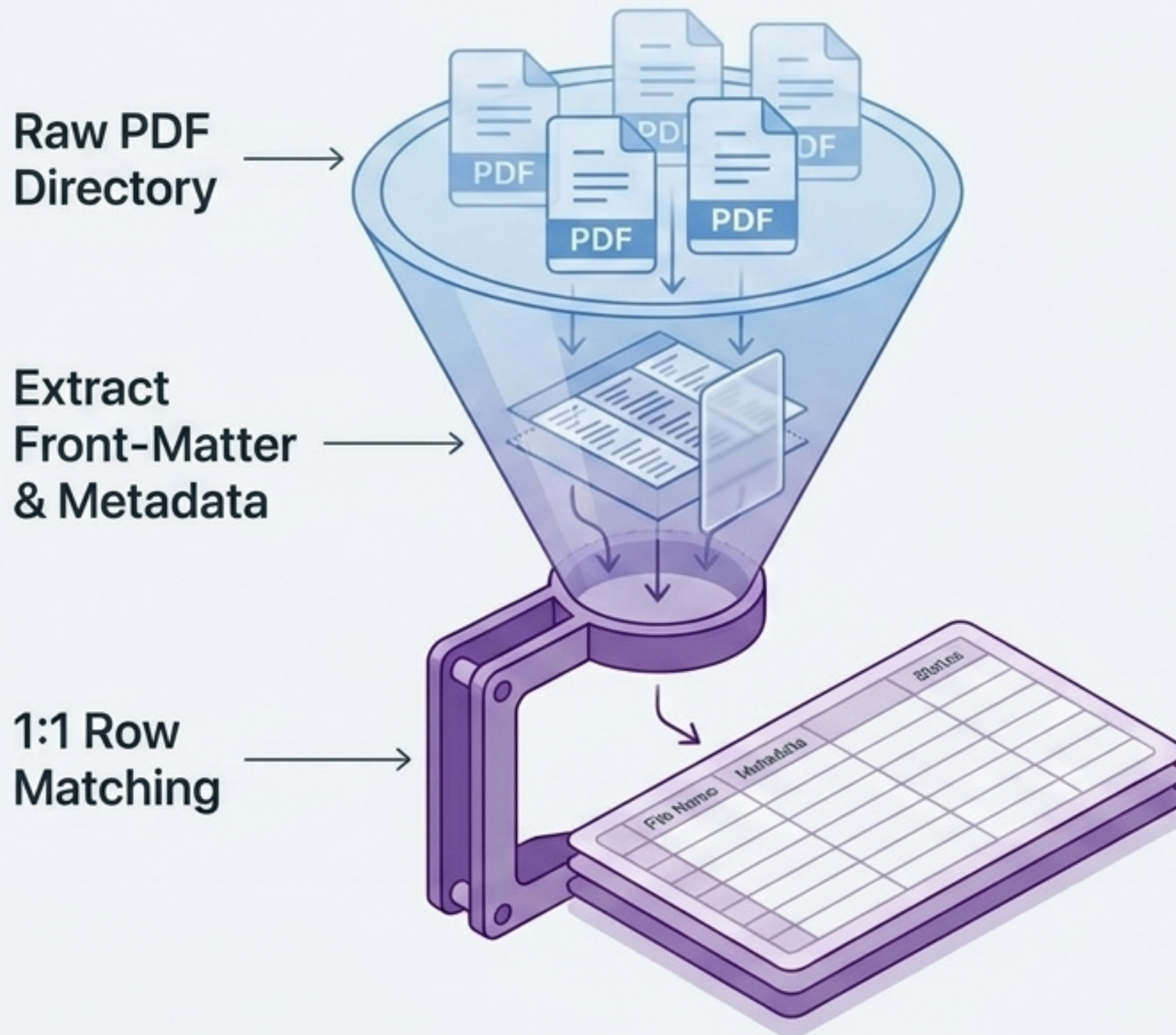


The Transparent Curation Pipeline



Auditable Data Flow

Parsing & Precision Matching



Strict 1:1 Matching

Each PDF aligns to exactly one spreadsheet row.

Surfaces Anomalies Early

Explicitly flags unmatched, ambiguous, and duplicate-row situations before extraction begins.

Targeted, Row-Aware Retrieval

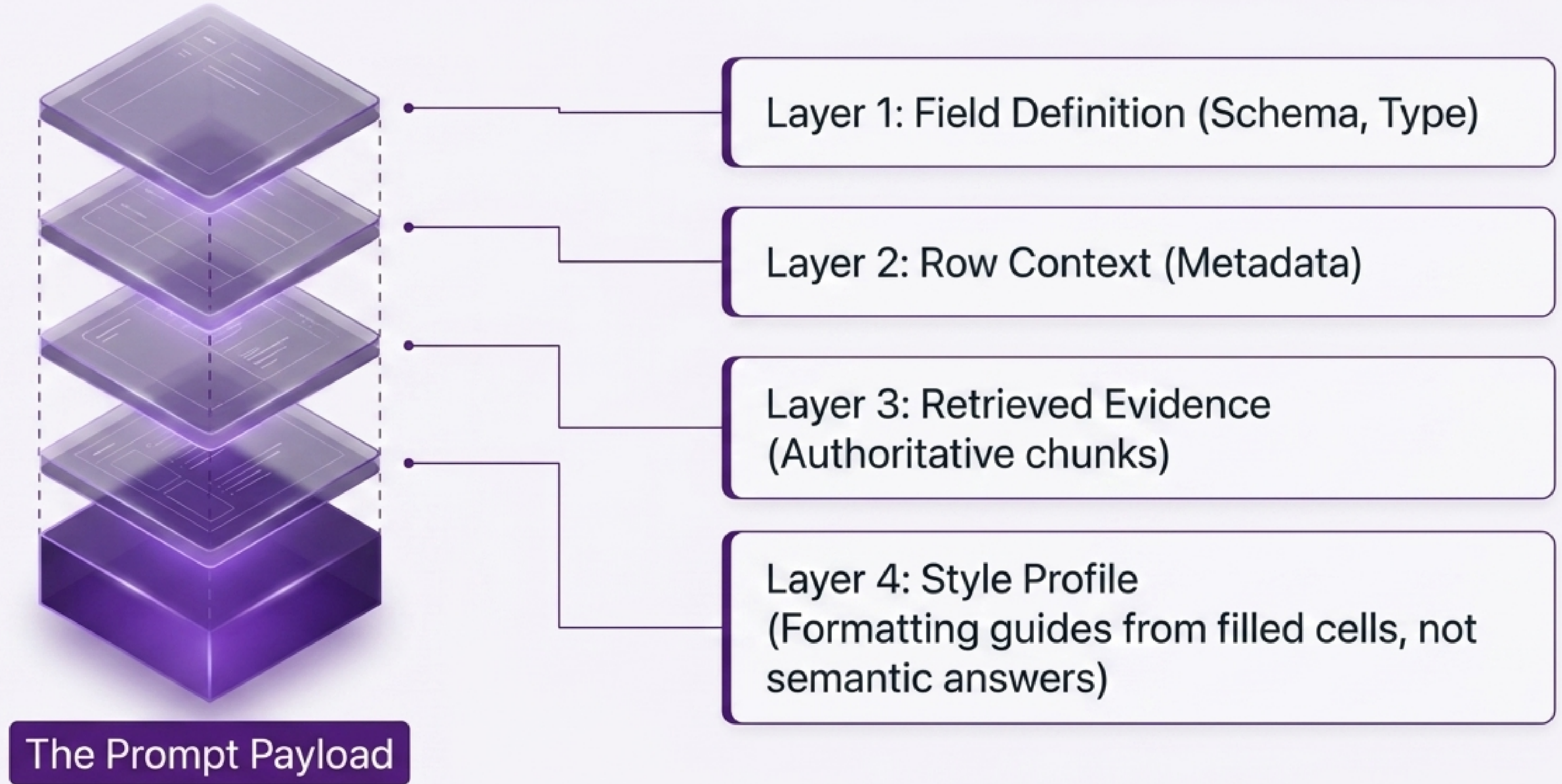


No Whole-Document Prompting: Avoids flooding the context window with irrelevant pages.

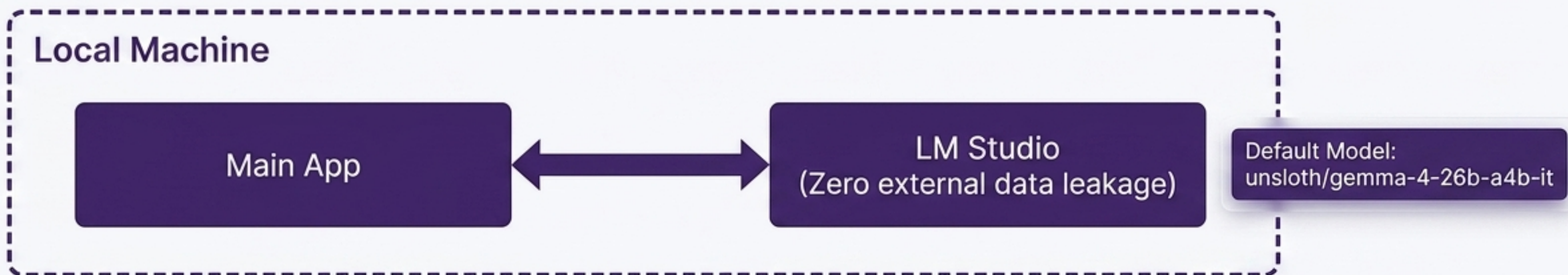
Precision Targeting: Uses row-aware and column-aware retrieval.

Default Engine: hybrid_experimental with top_k=12 to isolate only the most authoritative, field-relevant evidence.

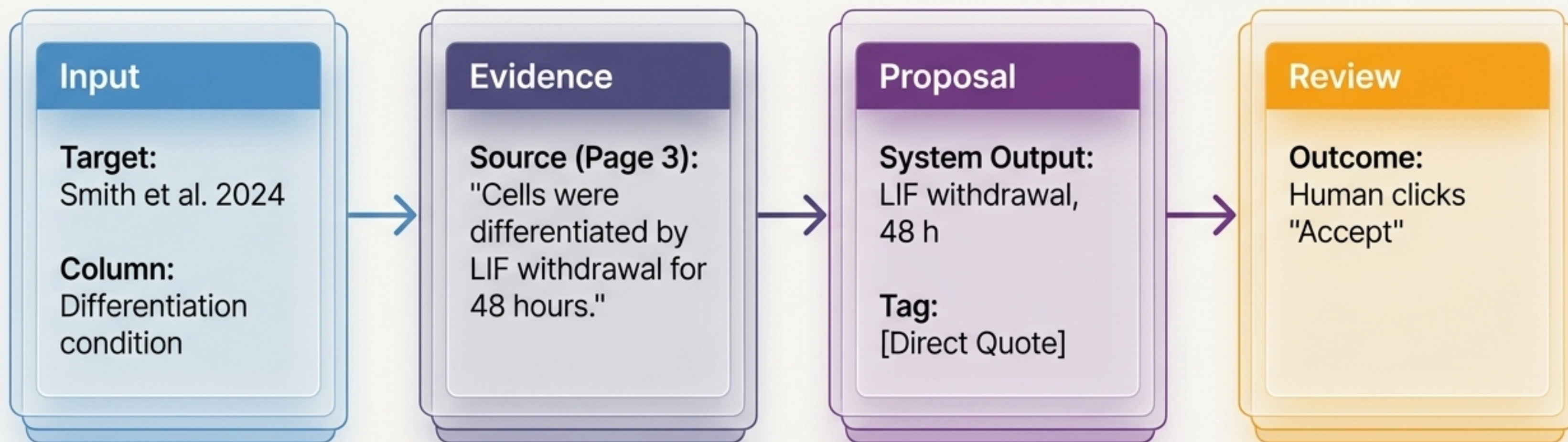
Exploded View: Prompt Composition



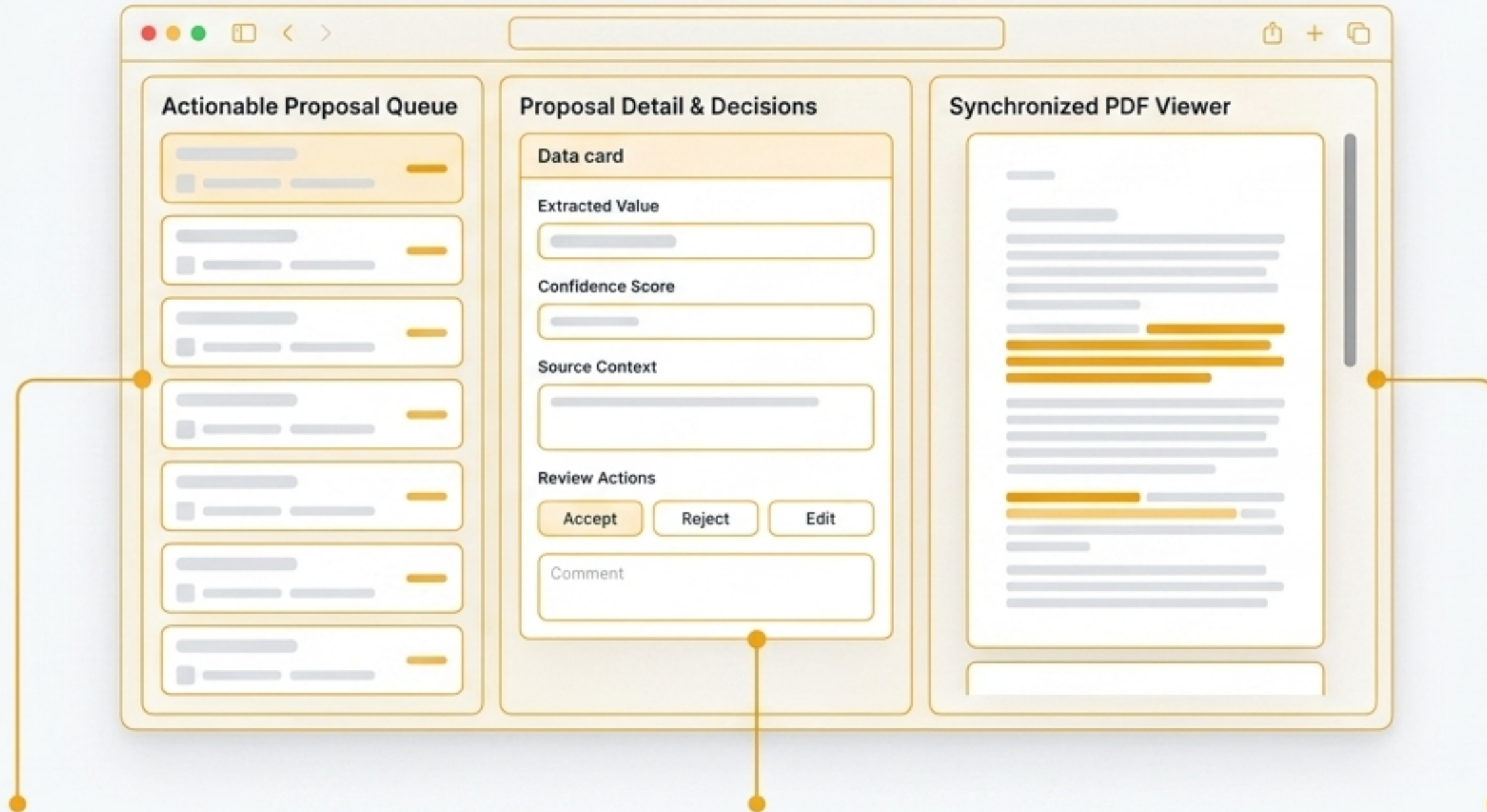
The Local-First Engine



The Journey of a Cell: A Worked Example



The Queue-First Review Workspace



Actionable Proposal Queue:

Grouped items optimized for rapid triage.
Excludes blocked/diagnostic clutter.

Proposal Detail & Decisions:

The active workspace for the current cell,
housing exact values and review actions.

Synchronized PDF Viewer:

The source document, automatically scrolled
and anchored to the exact extracted text.

Anatomy of a Review Decision

Explicit First-Class Curation

Value Block

Proposed Value with Support Marker distinguishes exact text from reasoning.

Decision Buttons

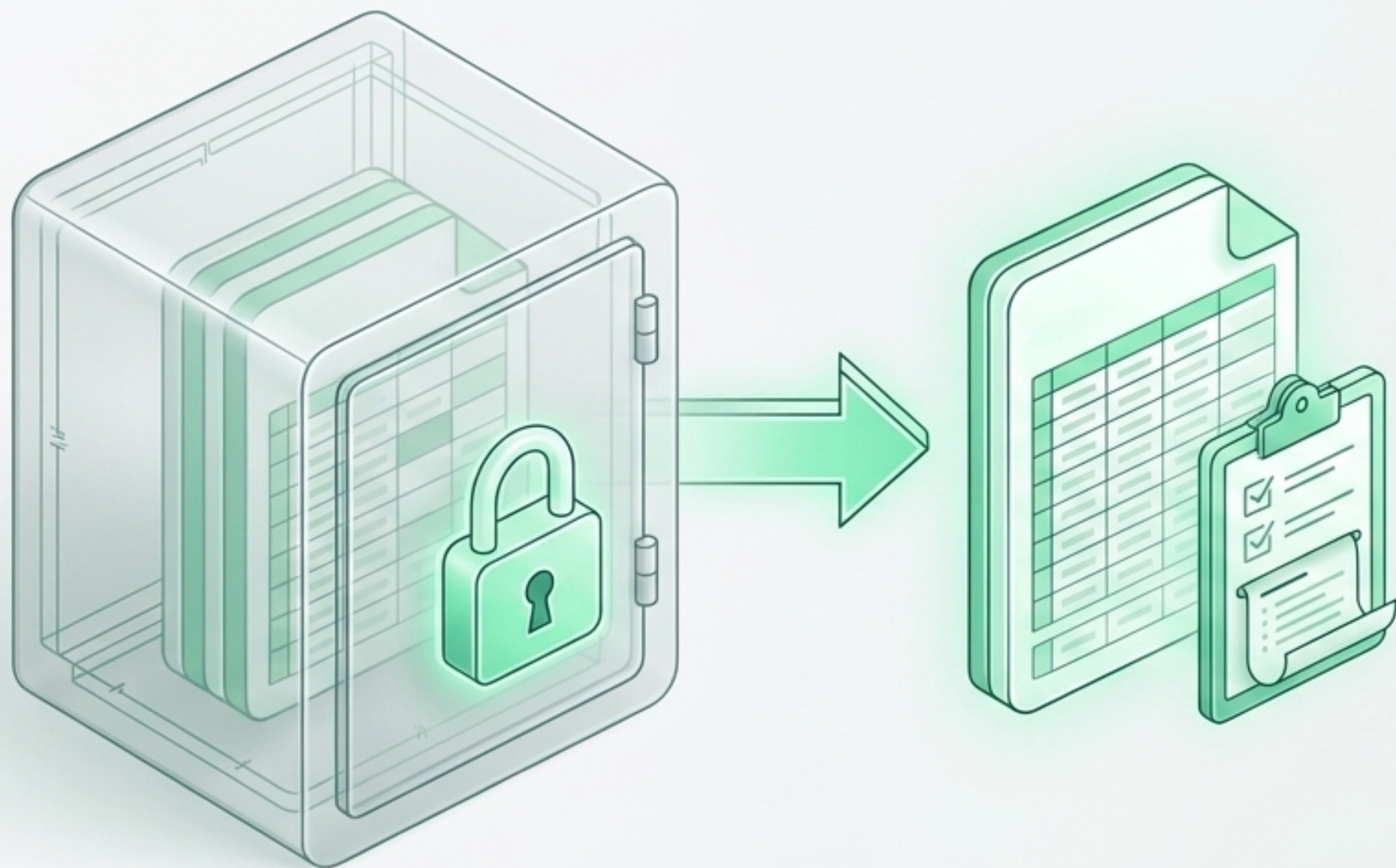
1. **Accept:** Validate as proposed.
2. **Edit:** Manually adjust the value before accepting.
3. **Confirm No Data:** Explicitly record that the paper lacks this information.
4. **Reject:** Discard the proposal entirely.

LIF withdrawal, 48 h

Direct Quote

Accept Edit Confirm No Data Reject

Auditable Export Discipline



- **Source Protection:**
Review decisions never mutate the source workbook in place.
- **Explicit Commit:**
Export bundles include only **explicitly accepted changes**. Unreviewed proposals are never exported as facts.
- **Audit Artifacts:**
Every exported cell is paired with a **transparent log** detailing the source paper, exact evidence snippet, and the specific human review decision.

Run Modes & Provenance



Normal Mode

Standard extraction for empty target cells.

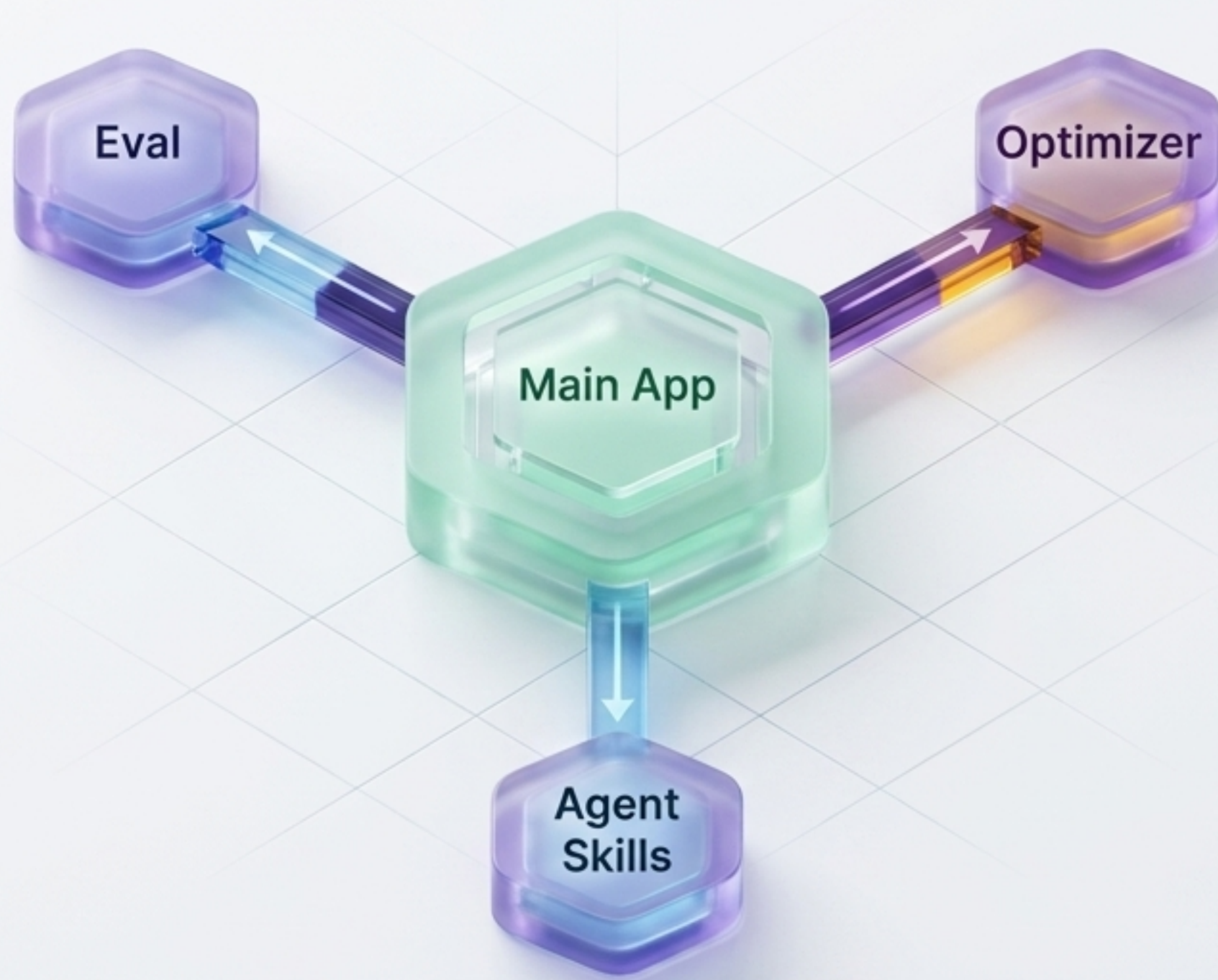
Verify Mode

Reviewer comparison on already-filled cells (validating existing data).

Eval Mode

Masked benchmark runs. Utilizes a protected working copy to ensure zero target-cell leakage during scoring.

The Monorepo Ecosystem



Center Hub: The **Main App** (The primary product: extraction, review, export).

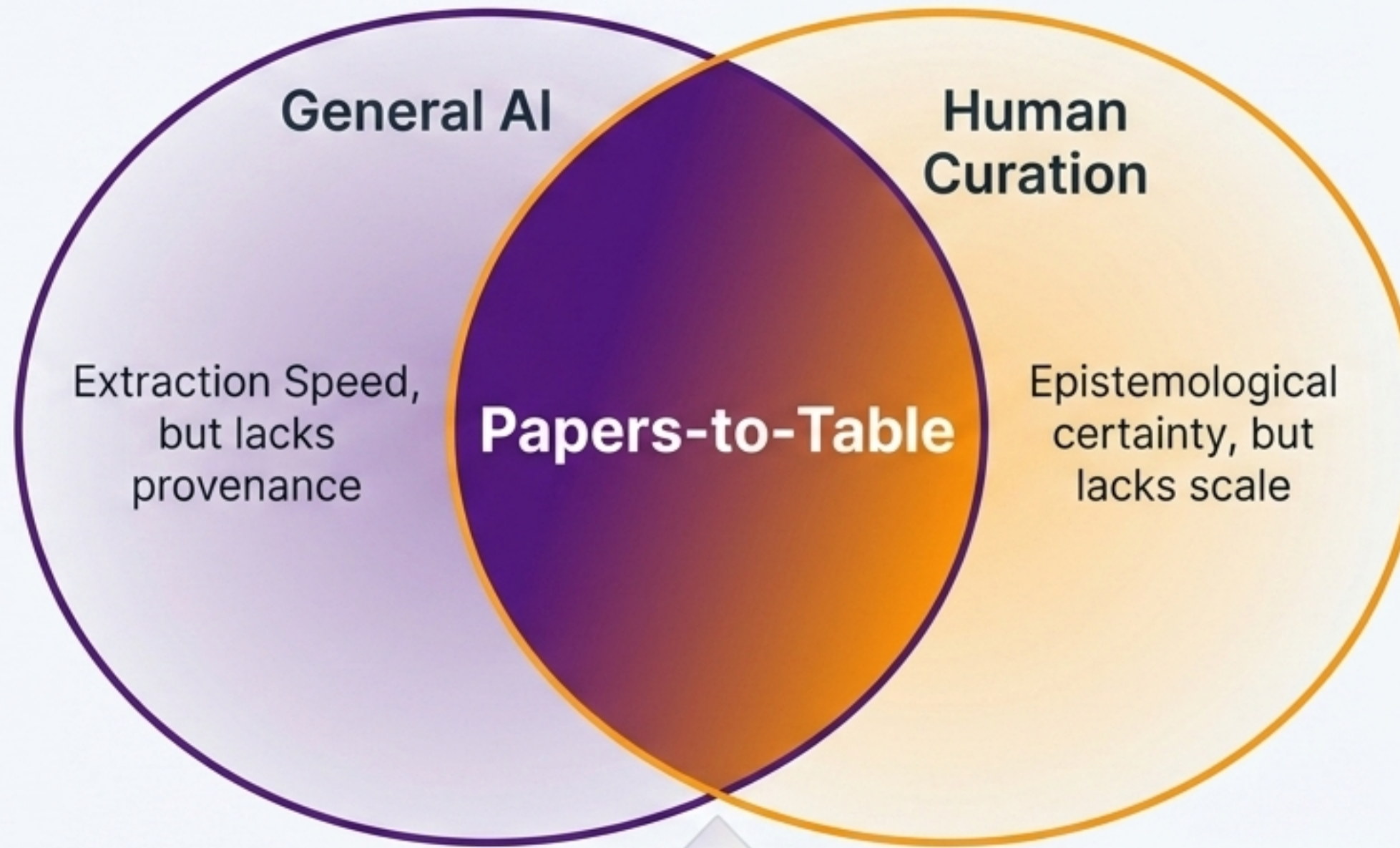
Satellite 1: **Eval** scores main-app output against human-verified benchmark data.

Satellite 2: **Optimizer** orchestrates candidate studies across different models and configs.

Satellite 3: **Agent Skills** are pluggable extraction skills for external systems.

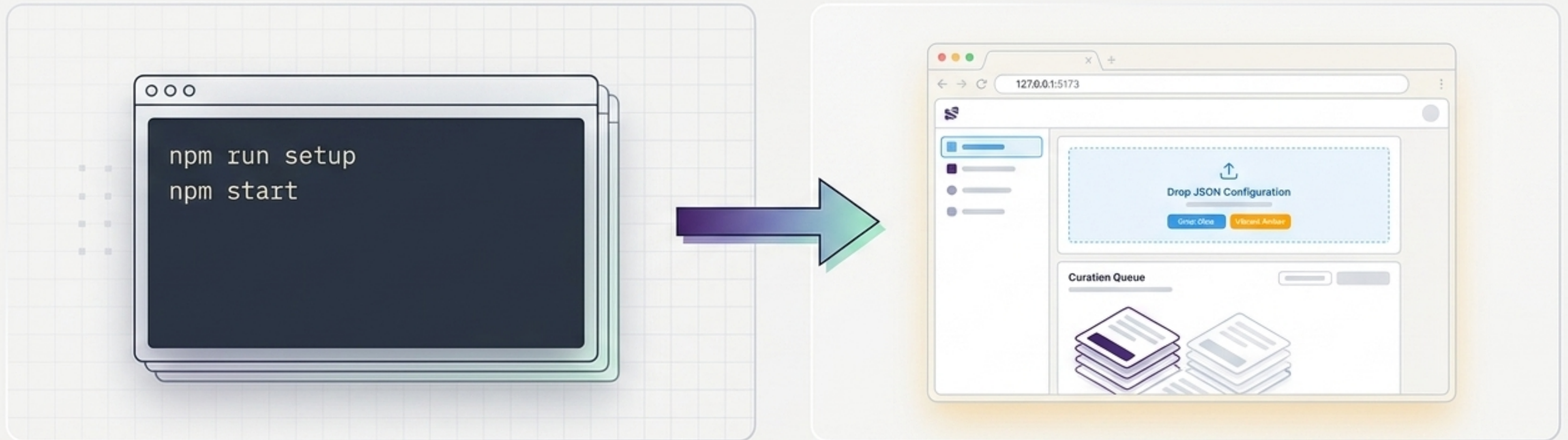
Rule: Companion tools consume auditable run bundles from files alone; they do not redefine the product.

The Exact Intersection



Papers-to-Table hits the precise synthesis: the extraction speed of AI combined with the epistemological certainty of human-reviewed evidence.

Deploy Locally. Curate with Certainty.



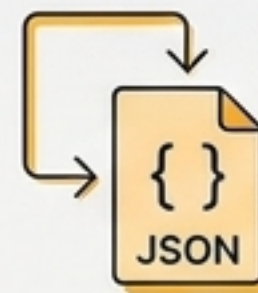
Next Step

1. Access the local-first browser app.



Next Step

2. Drop in a JSON configuration file.



Next Step

3. Begin your queue-first curation workflow.

